

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 2_COD_Question 4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ravi is developing a student registration system for a college. To efficiently store and manage the student IDs, he decides to implement a doubly linked list where each node represents a student's ID.

In this system, each student's ID is stored sequentially, and the system needs to display all registered student IDs in the order they were entered.

Implement a program that creates a doubly linked list, inserts student IDs, and displays them in the same order.

Input Format

The first line contains an integer N the number of student IDs.

The second line contains N space-separated integers representing the student IDs.

Output Format

The output should display the single line containing N space-separated integers representing the student IDs stored in the doubly linked list.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: 10 20 30 40 50

Answer

```
// You are using GCC
```

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
typedef struct Node{  
    int data;  
    struct Node* next;  
    struct Node* prev;  
} Node;
```

```
void insertend(Node** head,int data){  
    Node* newnode=(Node*)malloc(sizeof(Node));  
    newnode->data=data;  
    newnode->next=NULL;
```

```
    if(*head==NULL){  
        newnode->prev=NULL;  
        *head=newnode;  
        return;  
    }
```

```
    Node* temp = *head;  
    while(temp->next!=NULL){  
        temp=temp->next;  
    }
```

```
temp->next=newnode;
newnode->prev=temp;
}

void printlist(Node* head){
    Node* temp=head;
    while (temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}
```

```
int main(){
    int N,id;
    Node* head=NULL;
    scanf("%d",&N);

    for(int i=0;i<N;i++){
        scanf("%d",&id);
        insertend(&head,id);
    }
    printlist(head);
    return 0;
}
```

Status : Correct

Marks : 10/10